

Joint Distributions

It is defined as a probability distribution that assigns probabilities to combinations of two or more variables, capturing the likelihood of simultaneous occurrences of these variables.

Example:

										
Age (years):	7	7	7	8	8	9	9	9	9	10
Height (in):	45	46	46	47	47	49	49	49	50	50

		Height					
		45	46	47	48	49	50
Age	7	1	2	0	0	0	0
	8	0	0	2	0	0	0
	9	0	0	0	0	3	1
	10						

What is the probability that a child is 8 and 48 inches tall?

$$p_{XY}(x, y) = \mathbf{P}(X = x, Y = y)$$

$$p_{XY}(8, 48) = \mathbf{P}(X = 8, Y = 48)$$

$$p_{XY}(8, 48) = 0$$

$$p_{XY}(7, 46) =$$

		Height (Y)					
		45	46	47	48	49	50
Age (X)	7	1/10	2/10	0	0	0	0
	8	0	0	2/10	0	0	0
	9	0	0	0	0	3/10	1/10
	10	0	0	0	0	0	1/10

Probabilities

Joint Continuous Distributions

It happens when the X and y are continuous variable

X

Waiting time
before a call is picked up
[0 - 10 minutes]



2.4 minutes

1.5 minutes

Both variables are
continuous

 Y

Customer
satisfaction rating
[0 - 10]



0.0

5.7

X variable: Waiting time (mins)
0 - 10 mins

Y variable: Satisfaction rating
0 - 10

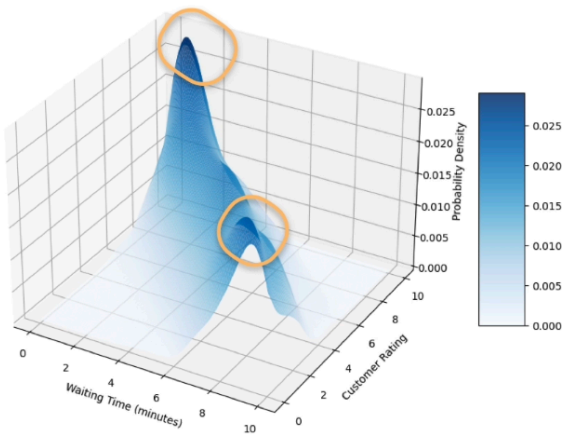
1000 customers

$\mathbb{E}[X] = 4.903$ minutes

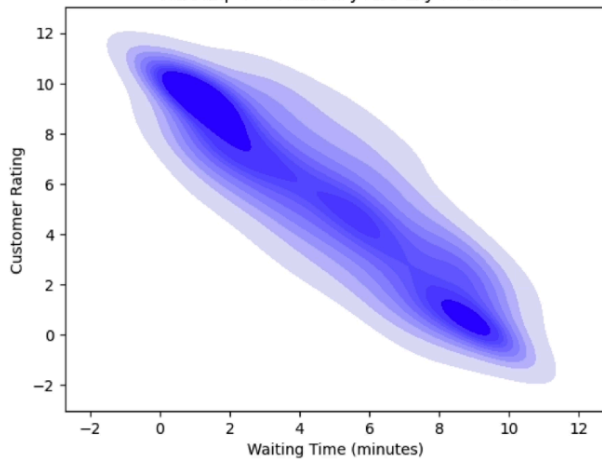
$\mathbb{E}[Y] = 5.280$



3D Probability Density Distribution for Customer Ratings vs Waiting Time



Heatmap of Probability Density Function

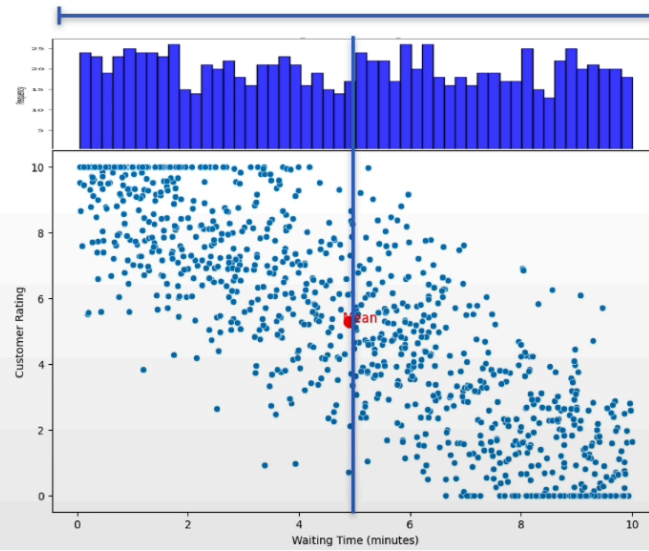


Variance

$$\mathbb{E}[X] = 4.903 \text{ minutes}$$

$$\mathbb{E}[X^2] = 32.561$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\ &= 32.561 - 4.903^2 \\ &= 8.526 \end{aligned}$$



$$\mathbb{E}[Y] = 5.280$$

$$\mathbb{E}[Y^2] = 38.037$$

$$\begin{aligned} \text{Var}(Y) &= \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \\ &= 38.037 - 5.280^2 \\ &= 10.163 \end{aligned}$$

